

AI-Powered Data Quality & Validation Framework

Whole Project Structure

(Complete)

AI-Data-Quality-Framework/



1. Client (Frontend)

client/

|

|─ public/

|

|─ src/

|

|

|─ assets/

|

|

|─ components/

|

|

Navbar.jsx

|

|

Sidebar.jsx

|

|

Loader.jsx

|

|

MetricCard.jsx

|

|

DataTable.jsx

|

|

UploadBox.jsx

|

|

SeverityBadge.jsx

|

|

ChartCard.jsx

|

|

|─ charts/

|

|

QualityChart.jsx

|

|

OutlierChart.jsx

|

|

MissingChart.jsx

|

|

CategoryChart.jsx

|

|

|─ pages/

|

|

Home.jsx

|

|

Upload.jsx

```
| | Dashboard.jsx
| | Validation.jsx
| | Anomaly.jsx
| | Quality.jsx
| | AIExplanation.jsx
| | Reports.jsx
| |
| |─ services/
| |   api.js
| |   uploadApi.js
| |   reportApi.js
| |
| |─ hooks/
| |
| |─ utils/
| |
| |─ context/
| |
| |─ App.jsx
| |
| |─ main.jsx
|
|─ package.json
|
|─ vite.config.js
```

2. Backend (Server)

Frontend pages

Landing Page



Upload Dataset



Dashboard



Validation Report



Anomaly Detection



Quality Score



AI Explanation



Download Reports

3. Server (Node & Express)

server/

|

├─ config/

|

db.js

|

multer.js

|

├─ controllers/

|

uploadController.js

|

validationController.js

|

reportController.js

|

dashboardController.js

|

├─ routes/

|

uploadRoutes.js

|

validationRoutes.js

|

reportRoutes.js

|

dashboardRoutes.js

|

├─ middleware/

|

auth.js

|

errorHandler.js

```
|
|
├── models/
|   User.js
|   Dataset.js
|   Report.js
|   Anomaly.js
|
|
├── services/
|   pythonService.js
|   reportService.js
|
|
├── uploads/
|
|
├── utils/
|
|
├── app.js
|
|
├── server.js
|
└── package.json
```

Backend APIs

POST

/api/upload

POST

[/api/validate](#)

POST

[/api/anomaly](#)

POST

[/api/report](#)

GET

[/api/dashboard](#)

GET

[/api/report/:id](#)

Flow of Node

React

↓

Upload CSV



Express



Save File



Python Script



Receive JSON



MongoDB



React Dashboard

3.ML

ml_engine/

|

└─ validation/

|

validator.py

|

missing.py

|

duplicate.py

|

datatype.py

|

date_validation.py

|

quantity_validation.py

|

price_validation.py

|

payment_validation.py

|

product_validation.py

|

status_validation.py

|

regex_validation.py

|

report.py

|

utils.py

|

└─ cleaning/

|

cleaner.py

|

missing_handler.py

|

date_cleaner.py

|

payment_cleaner.py

|

product_cleaner.py

|

price_cleaner.py

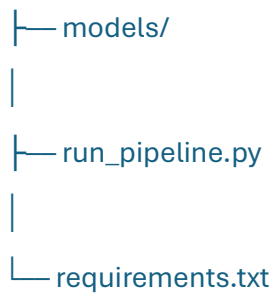
|

└─ rules/

|

business_rules.py

```
| rule_engine.py
|
|— statistics/
| zscore.py
| iqr.py
| outlier_engine.py
|
|— anomaly/
| isolation_forest.py
| local_outlier_factor.py
| dbscan_detector.py
| anomaly_engine.py
|
|— scoring/
| dataset_score.py
| record_score.py
| completeness.py
|
|— ai/
| gpt_summary.py
|
|— reports/
| pdf_report.py
| excel_report.py
| ppt_report.py
|
|— data/
```



ML Flow

CSV



Validation



Cleaning



Rule Engine



Statistical Detection



ML Detection



Quality Score



AI Summary



Generate Reports



Return JSON

Database

users

datasets

reports

anomalies

quality_scores

logs